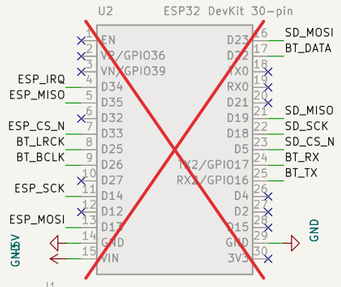
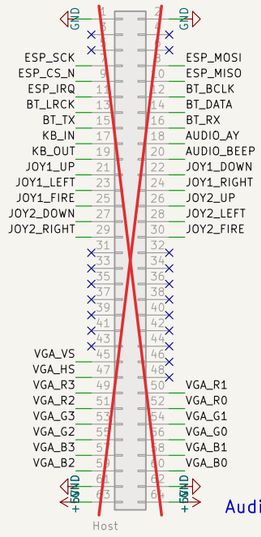
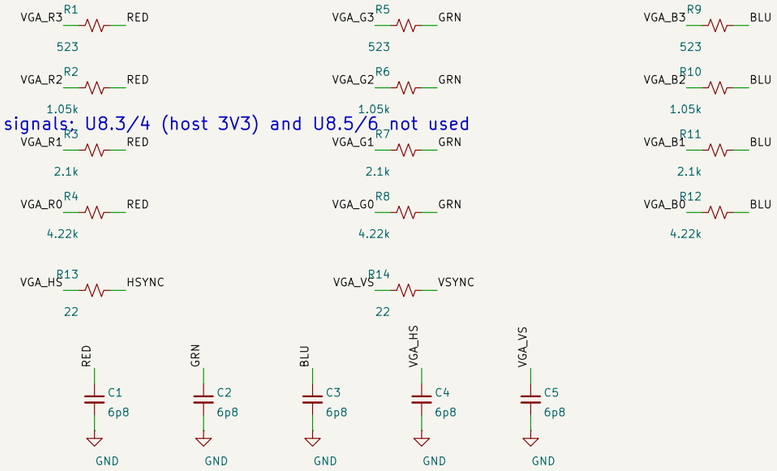
ESP32 system controller (SD card, tape loader, Bluetooth audio): dev board, 30-pin DOIT pinout, on 2x15x1 sockets; 3V3 pin not connected

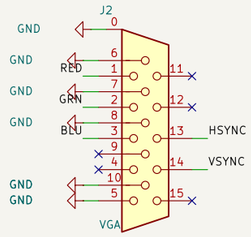


Host connector J1 = core board U8, pin for pin (stacks onto U8); 3.3 V signals; U8.3/4 (host 3V3) and U8.5/6 not used

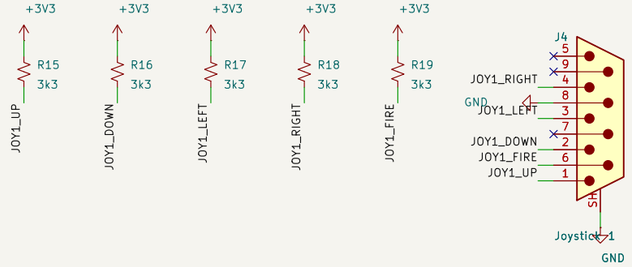


VGA: 4-bit DAC per colour (523/1.05k/2.1k/4.22k = 0.7 V full scale into 75 Ohm)

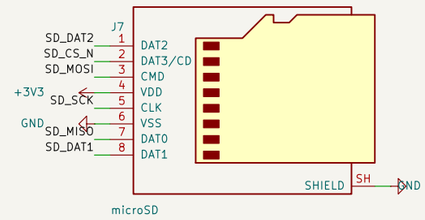




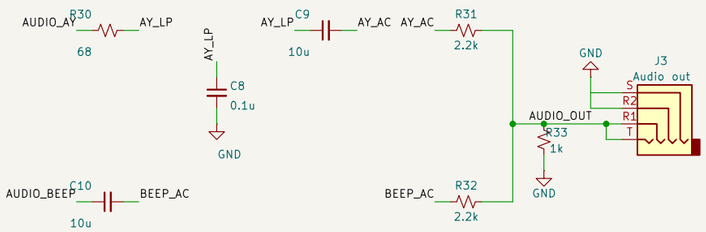
Joystick 1 (Atari/Kempston, right-angle DB9 male, pin 7 not connected)



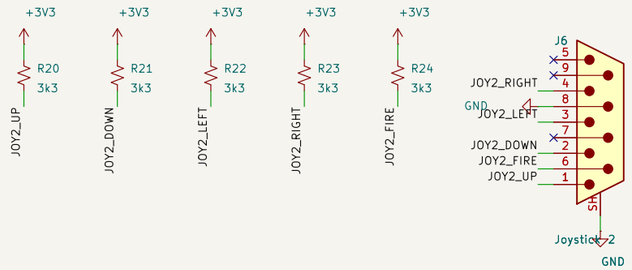
microSD (SPI, on the ESP32: D18 SCK, D19 MISO, D23 MOSI, D5 CS)

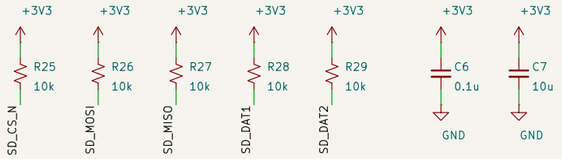


Audio out: AY + beeper to mono, 3.5 mm jack

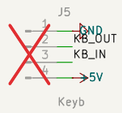


Joystick 2 (Atari/Kempston, right-angle DB9 male, pin 7 not connected)

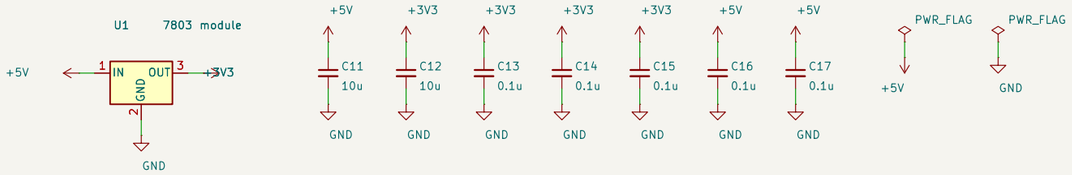




Keyboard module (USB HID -> UART, 3.3 V levels)



Power: 5 V in, 7803 switching module (SIP-3) -> 3.3 V



Mounting holes



4-bit VGA DAC, 2 joysticks, microSD, audio out (2 volume pots), keyboard module header
Separate board, wired to the QMTECH Cyclone IV board via J1

Sheet: /
File: Extra_PCB_CycloneIV.kicad_sch

Title: QM_Atrix Extended I/O board

Size: A3	Date: 2026-10-13	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 1/1